REST API Design Guidelines

Architecture board · AG-REST-02 · Version 1.4 · Effective 1 July 2026

> Fictional sample document for testing a domain-specialist RAG. Rules are cited in reviews as, for example, "API-07".

1. Resources and naming

API-01 Resource paths are plural nouns in kebab-case: `/v1/payment-methods`, never verbs such as `/getPayments`.

API-02 Every public API is versioned in the path (`/v1/`). Breaking changes require a new major version; the old version is supported for at least 12 months after the new one is released.

API-03 JSON field names use snake_case. Timestamps are ISO 8601 in UTC with a Z suffix. Money is an integer in minor units plus an ISO 4217 currency code.

2. Requests

API-04 List endpoints use cursor pagination with `limit` (default 20, maximum 100) and `cursor`. Offset pagination is not allowed on tables that can grow beyond 10,000 rows.

API-05 POST requests that create resources or move money require an `Idempotency-Key` header.

API-06 Validate every request body with a schema and reject unknown fields with HTTP 422.

3. Responses and errors

API-07 Errors use the Problem Details format (RFC 9457): `type`, `title`, `status`, `detail` and a `trace_id`. Never return stack traces, SQL or internal hostnames to clients.

```
{"type": "https://errors.example/validation", "title": "Invalid request",
 "status": 422, "detail": "amount must be positive", "trace_id": "4bf92f3577b34da6"}
```

API-08 Use the correct status codes: 201 for created, 202 for accepted asynchronous work, 204 for no content, 404 for missing resources, 409 for conflicts, 429 for rate limits with a Retry-After header.

4. Performance and limits

API-09 Every endpoint has a server-side timeout of 10 seconds and returns 504 if it is exceeded. Work that can take longer is asynchronous and returns 202 with a status URL.

API-10 Public endpoints are rate limited per API key; the default is 100 requests per minute.

API-11 Responses larger than 1 MB must be paginated or streamed. Avoid N+1 queries: list endpoints must fetch related data in a bounded number of queries.